Ayaan Hafeez

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Education

University of Alberta Edmonton, AB
BSc in Computer Engineering, Co-op, Software Option *Expected May 2028*
- **Relevant Coursework:** Data Structures & Algorithms, Software Requirements, Database Management (in progress)

Experience

AI Developer (3 co-op terms, then part-time) May 2025 – Aug 2025, Jan 2026 – Aug 2026 (co-op)
Elev8AI – Remote *Sep 2026 – Present (part-time)*
- Built an NL2SQL chatbot with LangChain, Ollama, ChromaDB, and RAG; 85% accuracy on internal benchmarks
- Built a full-stack social media platform: FastAPI, PostgreSQL, Redis, JWT auth, Stripe, React, TypeScript
- Built internal tools and Python automation: geospatial lookups, report pipelines, a customized Kimai fork
- Debugged and redeployed a production email-reporting job; migrating 170 NetSuite scripts to SuiteScript 2.1
- Turned customer meetings into requirements and data models for AI and ERP projects; ran end-to-end QA

Open Source

Omarchy – PR #8787 (merged) | *Arch + Hyprland Linux distro, 45K stars* Oct 2026 – Oct 2026
- Fixed a lopsided hover highlight in the network panel and pixel-snapped action icons at fractional scales

Vorssaint – PR #624 (merged) | *Swift, Core Graphics; macOS utility, 29K stars* Aug 2026 – Aug 2026
- Fixed display recovery that could leave a laptop with no usable screen after unplugging a monitor; added tests

Hyprspace – PR #4, PR #5 (merged) | *Swift; tiling window manager for macOS* Sep 2026 – Sep 2026
- Added smart single-window gaps, floating-window resize and centering keys, and modifier-drag to move windows

Projects

Trust Issues – NATHacks 2026, team Whats Your GPS (4 people) | *C, Go, MAVLink, ESP32* Nov 2026 – Nov 2026
- Built a lost-link detector that tells RF jamming from plain signal loss using heartbeats and radio RSSI
- In ArduPilot SITL it flagged jamming in 10.5 s and link loss in 5.2 s, then returned and landed at home
- Wrote the autonomy layer that sends EKF failover, hold, return, and land commands to the drone over MAVLink
- Built the Go/WebAssembly operator dashboard with a live MAVLink bridge; ported the core loop to an ESP32

surgraft – Token-Efficient Code Moves for LLM Tools | *Python, AST, Claude API, MCP* May 2026 – Jul 2026
- Built a CLI, Python library, and MCP server that moves functions between files for LLM coding tools
- Finds function bounds with Python's AST and byte-copies the exact range instead of having a model rewrite it
- Moving a 20-line function out of a 5,000-line file costs about 180 input tokens instead of about 44,000
- Sends only the moved snippet to Claude for an optional edit pass; published on PyPI

Kitsune – Keyboard Launcher for macOS | *Swift, AppKit, Lua 5.4, Unix sockets* Aug 2026 – Nov 2026
- Built a macOS launcher that opens on a global hotkey to fuzzy-search apps, commands, and system actions
- AppKit owns the native panel, app index, and IPC; an embedded Lua runtime owns menus, plugins, and themes
- Configs and plugins reshape it without touching Swift, and a kitsunectl CLI drives it from scripts
- Tag pushes publish releases to a Homebrew cask

ssht – Persistent SSH Sessions | *Rust, tokio, ratatui, SQLite, tmux, mosh* Jun 2026 – Aug 2026
- Built a Rust TUI that turns each SSH connection into a tmux session that survives sleep and network drops
- Wraps the system ssh instead of reimplementing it, so ProxyJump, keys, and hardware tokens keep working
- Hand-wrote an SSH config parser (Include recursion, Match blocks, wildcards); tokio probes hosts concurrently
- Fuzzy host picker with SQLite history and optional Mosh transport; published on crates.io and Homebrew

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL, Go, C, Rust, Swift
Backend: FastAPI, PostgreSQL, Redis, SQLite, REST APIs, JWT auth, Stripe
Frontend: React, Vite, Tailwind CSS, WebExtensions API
AI / LLM: LangChain, Ollama, ChromaDB, RAG, Claude API, OpenAI API, MCP
Tools: Git, Docker, nginx, Linux, Homebrew